

Experiment 8 — LIME

Explainability of a Pre-trained Image Classification Model

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Code Repository: https://github.com/CodeCraftsmanRaj/xAI/tree/main/Experiment_8_LIME

1. Introduction

LIME (Local Interpretable Model-agnostic Explanations) is a post-hoc explainability technique that approximates a complex model locally with an interpretable surrogate model. For image classification, LIME divides an image into superpixels, creates perturbed versions by hiding selected regions, observes the model's predictions, and assigns importance weights to regions associated with a selected class.

This experiment applies LIME to a pretrained ResNet-50 ImageNet classifier. Three dog images with different backgrounds were evaluated. The experiment then varied SLIC segmentation resolution to create two reasonable but substantially disagreeing explanations for one study image. Finally, deletion and insertion tests used only the model's prediction probabilities to compare the competing explanations.

2. Aim

To explain prediction of a pre-trained model using LIME.

3. Objectives

- 1. Apply LIME to three images and visualize important regions.
- 2. Compare explanations under different SLIC superpixel counts.
- 3. Identify two reasonable but disagreeing explanations.
- 4. Define the decision criterion before testing.
- 5. Use only model predictions through deletion and insertion tests.
- 6. Determine whether the evidence supports A, B, or neither.

4. Experimental Setup

Component	Configuration
Model	ResNet-50
Pretraining	ImageNet
Device	CUDA / GPU
LIME samples	1200
LIME features	12
Segmentation	SLIC
Base segments	80
Candidate segments	40, 80, 120
Study image	dog1.jpg
Target class	golden retriever
Test fractions	10%, 20%, 30%, 40%, 50%
Baseline	Blur
Decision margin	0.05

5. Methodology

5.1 Model prediction

Each image was passed through pretrained ResNet-50 and the top ImageNet predictions and probabilities were recorded.

5.2 LIME attribution

LIME generated superpixel-level explanations using 1200 perturbed samples. Positive regions were used to represent regions supporting the target prediction; local surrogate R^2 was recorded as fidelity.

5.3 Disagreement experiment

For dog1.jpg, SLIC segmentation was varied while the other major settings were kept fixed. Candidate segment counts were 40, 80, and 120. The selected pair was 40 vs 120 segments.

5.4 Pre-registered decision criterion

A would be supported if its positive regions caused a larger target-probability decrease during deletion and/or retained a larger target probability during insertion than B, with the combined difference exceeding 0.05. B would be supported by the reverse pattern. Otherwise the result would be neither/inconclusive.

5.5 Prediction-only validation

Positive regions were progressively deleted from the image and progressively inserted into a blurred baseline. Only the model's target-class probability was used; no ground-truth segmentation was used.

6. Predictions and LIME Results

Image	Class	Probability	LIME R^2	Segments	Samples
dog1.jpg	golden retriever	0.5142	0.6093	80	1200
dog2.jpg	golden retriever	0.4172	0.4979	80	1200
dog3.jpg	golden retriever	0.3979	0.6798	80	1200

All three images were classified as golden retriever. Probabilities were 0.5142, 0.4172, and 0.3979.

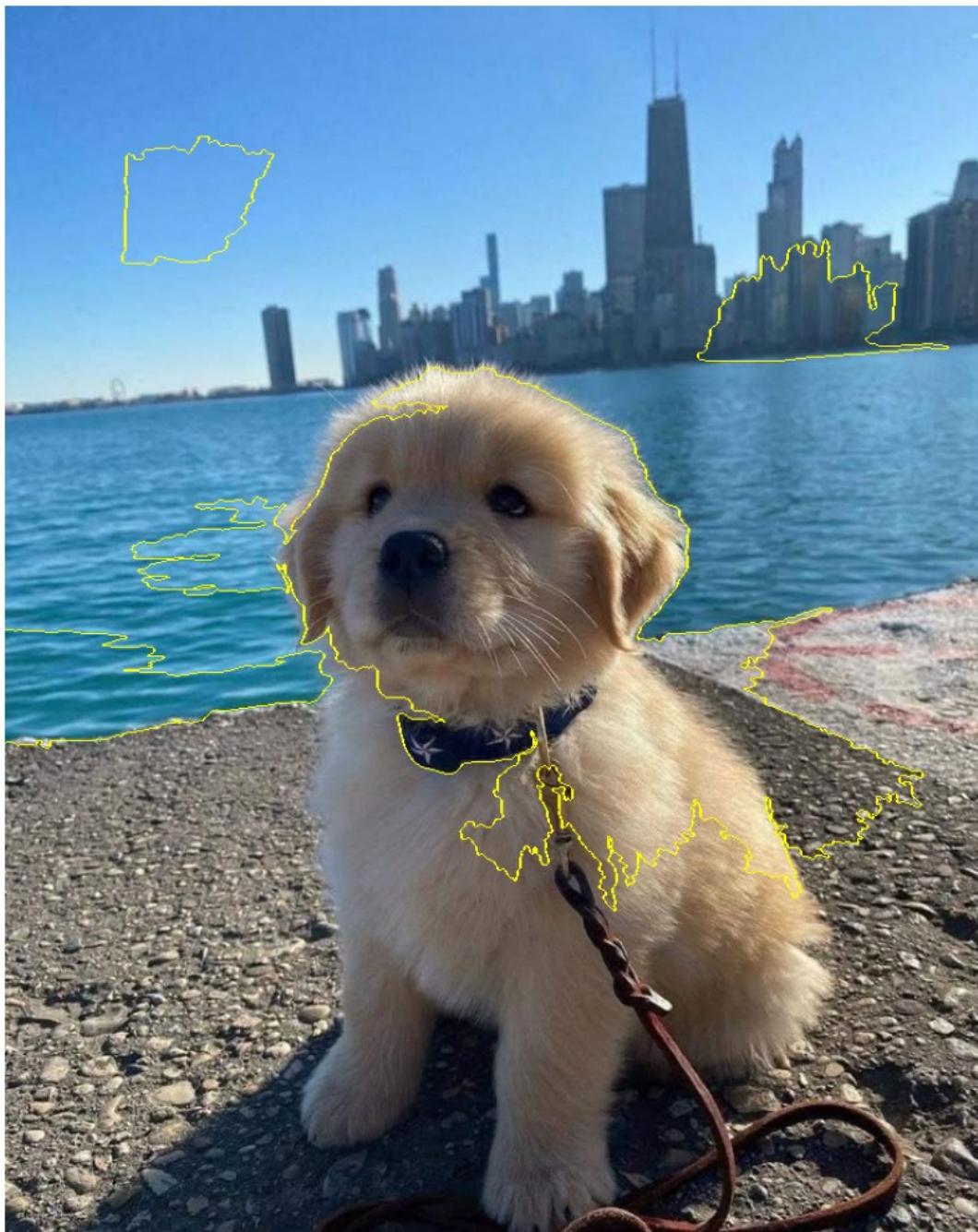
dog1.jpg — LIME attribution

LIME positive attribution — golden retriever



dog2.jpg — LIME attribution

LIME positive attribution — golden retriever



dog3.jpg — LIME attribution

LIME positive attribution — golden retriever



7. Disagreement Analysis

A seg.	B seg.	A R ²	B R ²	Jaccard	Disagreement	Reasonable
40	80	0.8242	0.6093	0.2659	73.41%	Yes
40	120	0.8242	0.5374	0.2519	74.81%	Yes
80	120	0.6093	0.5374	0.4903	50.97%	Yes

Selected pair: A = 40 segments, B = 120 segments. Pixel-mask disagreement = 74.81%; Jaccard overlap = 0.2519.

Explanation A — 40 segments

LIME positive attribution — golden retriever



Explanation B — 120 segments

LIME positive attribution — golden retriever



Direct comparison of A and B

Disagreement comparison — golden retriever

Explanation A



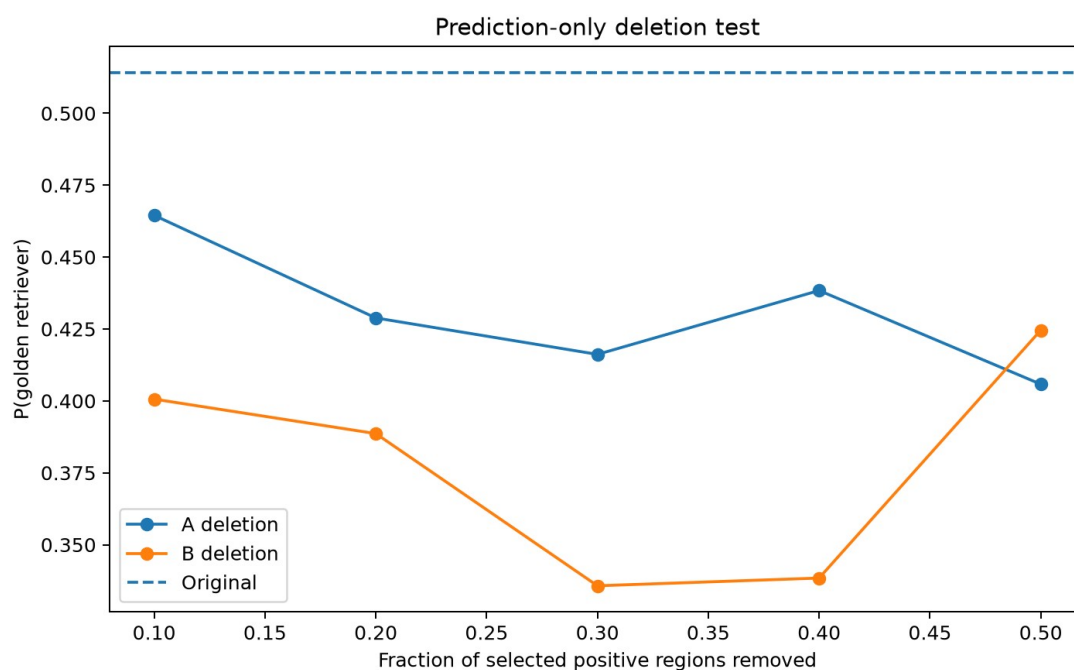
Explanation B



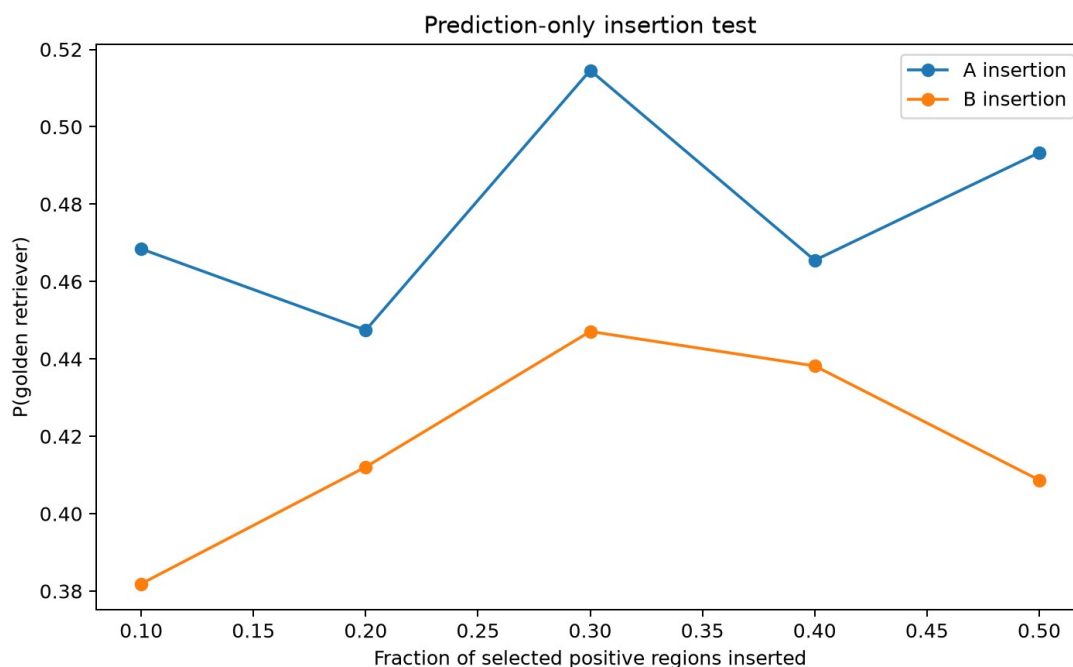
8. Prediction-Only Validation

Metric	A	B
Deletion AUC (probability drop)	0.033839	0.058156
Insertion AUC	0.190840	0.169268
Combined A – B support	-0.002746	—
Decision margin	0.050000	—
Final decision	Neither / inconclusive	

Deletion test



Insertion test



9. Detailed Prediction-Test Results

Fraction	A deletion P	B deletion P	A insertion P	B insertion P
10%	0.4645	0.4006	0.4684	0.3819
20%	0.4288	0.3886	0.4474	0.4121
30%	0.4162	0.3357	0.5146	0.4471
40%	0.4384	0.3384	0.4655	0.4382
50%	0.4058	0.4246	0.4933	0.4087

10. What Is Understood from the Results

Consistent classification

All three dog images received the same predicted class, golden retriever, although confidence varied from 39.79% to 51.42%.

Sensitivity to segmentation

Changing the SLIC segmentation produced substantially different highlighted regions. The selected 40-vs-120 pair disagreed over 74.81% of the evaluated pixel mask.

Fidelity is not proof

A had higher local surrogate fidelity (R^2 0.8242) than B (0.5374), but this alone does not establish that A is the correct explanation.

Prediction-only result

B had the larger deletion probability drop, while A had the larger insertion score. Because the signals conflict and the combined difference is only -0.002746, below the 0.05 margin, neither explanation is supported strongly enough.

XAI lesson

Explanation sensitivity is itself an important result. When reasonable explanations disagree, a validation procedure should be used rather than selecting a visualization by appearance alone.

11. Limitations

- The model is pretrained on ImageNet rather than trained specifically on these images.
- The three predictions have moderate confidence.
- LIME is local and can vary with segmentation, sampling, or random seed.
- Prediction-only validation measures consistency with model behavior, not real-world causal truth.
- Perturbation baselines can affect deletion/insertion measurements.
- The model may use contextual or spurious visual cues.

12. Conclusion

The experiment successfully applied LIME to a pretrained ResNet-50 ImageNet classifier and generated attribution maps for three dog images. All three were classified as golden retriever. For dog1.jpg, changing SLIC segmentation from 40 to 120 segments produced two reasonable but strongly disagreeing explanations, with 74.81% pixel-mask disagreement.

The pre-defined prediction-only validation did not establish a winner. Explanation B produced the larger deletion probability drop, while explanation A produced the larger insertion score. The combined support difference was -0.002746, far below the 0.05 decision margin. Therefore, the experimental conclusion is neither/inconclusive.

The main lesson is that LIME explanations are local approximations and can be sensitive to methodological choices. A visually plausible explanation should not automatically be treated as ground truth; when explanations disagree, explicit validation is valuable.

13. Reproducibility and Repository

Code: https://github.com/CodeCraftsmanRaj/xAI/tree/main/Experiment_8_LIME

Generated outputs include predictions.csv, summary.json, decision.md, decision_criterion.md, three_image_lime.csv, disagreement_pairs.csv, prediction_test.csv, attribution maps, explanation comparisons, and deletion/insertion plots.

14. Key Takeaways

- LIME provides useful local visual explanations for image classifiers.
- Explanations can change substantially with segmentation granularity.
- Local fidelity should not be interpreted as proof of causal correctness.
- Deletion and insertion tests can compare explanations using only model predictions.
- When validation evidence is conflicting or too small, an inconclusive result is scientifically preferable to forcing a winner.